

Foundations of Arithmetic Power Geometry VIII: A Spectral–Green Bridge for APG Energy and the Boundary-Control Program

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Abstract

This paper develops the eighth volume of Arithmetic Power Geometry (APG) by proving an unconditional analytic bridge between APG source distributions, Poisson potentials, Green energy, and spectral control on compactified modular curves. Earlier APG volumes introduced local entropy-governed closure defects, information-geometric stabilization, integrated defect energy, scale-corrected invariants, discrete prime-chain regularization, and a conditional APG–Arakelov projection program. The remaining difficulty identified in APG VI–VII is the conversion of continuous APG deformation energy into conductor-scale Arakelov intersection control. This paper proves a Spectral–Green Bridge Theorem: on the compact Riemann surface

$$X = X_0(N)^*(\mathbb{C}),$$

the APG Dirichlet energy of the mean-zero Poisson potential satisfies

$$E_{\text{APG}} \leq \lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2,$$

where $\lambda_1(\Delta_{\text{APG}}) > 0$ is the first nonzero eigenvalue of the positive APG Laplacian. The theorem is unconditional once the APG metric is fixed as a smooth compact Arakelov-compatible metric. We then formulate the Spectral Conductor Criterion, showing that the Green-energy distortion estimate required in APG VII reduces to an explicit spectral-conductor inequality. The paper does not claim a proof of Fermat’s Last Theorem, the *abc* conjecture, the Szpiro conjecture, or the full APG–Arakelov Projection Theorem. Its contribution is a rigorous analytic reduction: the APG projection wall is transformed into a precise spectral bound involving the first Laplace eigenvalue and the L^2 -size of the APG source.

Keywords: Arithmetic Power Geometry; Arakelov geometry; modular curves; Laplacian; Green functions; spectral gap; Dirichlet energy; Poisson equation; Faltings height; APG projection theorem.

1 Introduction

Arithmetic Power Geometry (APG) studies algebraic closure under exponent deformation (Akhtar, 2026h). Instead of fixing an exponent and varying coordinates, APG fixes positive base coordinates and treats the exponent as a deformation parameter. The first volume established a local closure-defect expansion at the Euclidean target, where the leading coefficient is governed by Shannon entropy (Shannon, 1948; Cover & Thomas, 2006). Later volumes developed stabilized information geometry, integrated defect energy, scale correction, discrete prime-chain regularization, and a conditional bridge toward Arakelov intersection theory.

The main unresolved obstacle is the analytic-to-arithmetic projection problem. APG VI introduced a conditional APG–Arakelov height coupling program in which deformation energy is expected to influence the Faltings height of a Frey curve through modular parametrization and Arakelov self-intersection. APG VII sharpened the problem by reducing the projection bridge to explicit boundary estimates: weighted cusp-width control, vertical component multiplicity control, and Green-energy metric distortion control. Among these, the Green-energy distortion estimate is analytically the most accessible because it belongs to the theory of elliptic operators, Green functions, spectral gaps, and Dirichlet energies on compact Riemann surfaces.

This paper proves an unconditional analytic theorem inside the APG architecture. The theorem does not complete the full APG–Arakelov projection theorem, but it proves a necessary and rigorous bridge:

$$\rho_{\text{APG}} \longrightarrow u_{\text{APG}} \longrightarrow E_{\text{APG}} \longrightarrow \lambda_1(\Delta_{\text{APG}})^{-1} \|F_{\text{APG}}\|_2^2.$$

Here ρ_{APG} is the APG source distribution, F_{APG} is its mean-zero normalization, u_{APG} is the unique mean-zero Poisson potential, and E_{APG} is the corresponding Dirichlet energy. The new result is called the Spectral–Green Bridge Theorem.

The purpose of this paper is therefore precise. We do not introduce another speculative global invariant. Instead, we prove that APG energy is controlled by the spectral geometry of the compactified modular curve. This transforms the Green-energy obstruction of APG VII into a concrete spectral-conductor problem. The next stage of the APG program is no longer the creation of new formal objects, but the proof of explicit conductor-scale bounds for

$$\lambda_1(\Delta_{\text{APG}})^{-1} \|F_{\text{APG}}\|_2^2.$$

Relationship to Earlier APG Volumes

The present work constitutes the eighth volume of the Arithmetic Power Geometry (APG) research program. APG I established the foundational framework of exponent deformation and proved the first-order entropy-governed closure-defect asymptotic expansion near the Euclidean target (Akhtar, 2026a). APG II extended the framework through information-geometric formulations, introducing a stabilized metric structure, coordinate-stability principles, and logarithmic projective-height mappings (Akhtar, 2026b). APG III developed the integrated closure-defect functional A_0 and formulated the conditional Frey-curve height-obstruction program (Akhtar, 2026c). APG IV introduced scale-corrected defect invariants and established

the coordinate-dependent damping obstruction framework, clarifying the analytical limits of continuous deformation models (Akhtar, 2026d). APG V developed the discrete regularization architecture, replacing ill-defined continuous conductor trajectories with prime-chain formulations and scale-dependent arithmetic structures (Akhtar, 2026e).

Building upon these foundations, APG VI proposed a conditional APG–Arakelov height-coupling program, introducing the APG Poisson-energy framework and the modular-efficiency architecture linking deformation energy to arithmetic height theory (Akhtar, 2026f). APG VII subsequently established the discrepancy–source compatibility theorem and formulated the conditional APG–Arakelov projection-theorem architecture, reducing the analytic-to-arithmetic projection problem to explicit boundary estimates involving cusp widths, vertical components, and Green-energy distortion fields (Akhtar, 2026g).

APG 0 provides the unified axiomatic synthesis of the entire program, consolidating the fundamental objects, invariants, theorem hierarchy, examples, counterexamples, and open problems of Arithmetic Power Geometry into a single foundational framework (Akhtar, 2026h). The present paper develops the spectral component of that architecture by establishing a Spectral–Green bridge linking APG source distributions, Poisson potentials, Dirichlet energy, Green operators, and spectral control on compactified modular curves. In doing so, it transforms the principal Green-energy obstruction identified in APG VII into a precise spectral-conductor problem, thereby providing a rigorous analytical foundation for future developments of the APG–Arakelov projection program.

2 Background and Scope

Classical arithmetic geometry studies rational and integral points on algebraic varieties using heights, divisors, Galois representations, modular forms, and intersection theory (Bombieri & Gubler, 2006; Faltings, 1983, 1984; Hartshorne, 1977; Hindry & Silverman, 2000; Lang, 1983, 2002; Silverman, 1986, 1992, 2007; Diamond & Shurman, 2005; Katz & Mazur, 1985; Shimura, 1971). The proof of Fermat’s Last Theorem proceeds through the Frey curve, Ribet’s level-lowering theorem, and modularity lifting (Frey, 1986; Mazur, 1977; Ribet, 1990; Serre, 1987; Taylor & Wiles, 1995; Wiles, 1995).

APG does not replace this classical machinery. Its contribution is different: it supplies a deformation-energy language for measuring how algebraic closure changes when the exponent moves away from the Euclidean baseline $n = 2$. The APG program is compatible with known arithmetic geometry but does not by itself imply global Diophantine impossibility.

This paper has three boundaries. First, all analytic results are proved on the compact Riemann surface

$$X = X_0(N)^*(\mathbb{C}),$$

not on the noncompact modular curve before cusp compactification. This avoids boundary ambiguity in the Poisson equation. Second, the Laplacian is always the positive Laplacian

$$\Delta_{\text{APG}} = -\text{div}_{\text{APG}} \nabla_{\text{APG}}.$$

This sign convention ensures

$$\int_X u \Delta_{\text{APG}} u \, d\mu = \int_X |\nabla_{\text{APG}} u|^2 \, d\mu.$$

Third, the theorem proves analytic Green-energy control. It does not automatically prove the full arithmetic intersection inequality

$$|\widehat{c}_1(\overline{\omega}_X)^2 - \deg(\pi) \widehat{c}_1(\overline{\omega}_E)^2| \leq C \deg(\pi) \log N.$$

That estimate still requires boundary comparison between analytic APG potentials and Arakelov discrepancy metrics.

3 APG Source Normalization

Let $X = X_0(N)^*(\mathbb{C})$ be a compact connected Riemann surface equipped with a smooth APG–Arakelov-compatible metric g_{APG} . Let $d\mu$ be the associated volume form and let

$$\text{Vol}(X) = \int_X d\mu.$$

Let $\rho_{\text{APG}} \in C^\infty(X)$ be a nonnegative APG source distribution satisfying

$$\int_X \rho_{\text{APG}} \, d\mu = 1.$$

Define the mean-zero APG source field

$$F_{\text{APG}} = \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)}.$$

Then

$$\int_X F_{\text{APG}} \, d\mu = \int_X \rho_{\text{APG}} \, d\mu - \frac{1}{\text{Vol}(X)} \int_X d\mu = 1 - 1 = 0.$$

Thus F_{APG} lies in the mean-zero subspace $L_0^2(X)$.

4 The APG Laplacian

Define the positive APG Laplacian by

$$\Delta_{\text{APG}} = -\text{div}_{\text{APG}} \nabla_{\text{APG}}.$$

Since X is compact and g_{APG} is smooth, Δ_{APG} is a nonnegative self-adjoint elliptic operator on $L^2(X)$. Standard elliptic spectral theory on compact Riemannian manifolds implies that its spectrum is discrete (Buser, 1992; Jost, 2017; Rudin, 1976):

$$0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \cdots,$$

where $\lambda_0 = 0$ corresponds to constant functions. Because X is connected, constants are the only harmonic functions. Hence

$$\lambda_1(\Delta_{\text{APG}}) > 0.$$

Spectral gaps on modular and arithmetic surfaces play a central role in analytic number theory and automorphic forms (Sarnak, 1995).

5 The APG Poisson Problem

Because F_{APG} has mean zero, the Poisson equation

$$\Delta_{\text{APG}} u_{\text{APG}} = F_{\text{APG}}$$

has a solution unique up to constants. We impose the normalization

$$\int_X u_{\text{APG}} d\mu = 0.$$

This gives a unique mean-zero solution

$$u_{\text{APG}} \in C^\infty(X).$$

Define the APG Dirichlet energy by

$$E_{\text{APG}} = \int_X |\nabla u_{\text{APG}}|^2 d\mu.$$

This is the Green energy of the APG source.

6 Main Theorem: The Spectral–Green Bridge

Theorem 6.1 (Spectral–Green Bridge Theorem). *Let $X = X_0(N)^*(\mathbb{C})$ be a compact connected Riemann surface equipped with a smooth APG–Arakelov-compatible metric. Let Δ_{APG} be the positive Laplacian. Let $\rho_{\text{APG}} \in C^\infty(X)$ satisfy*

$$\int_X \rho_{\text{APG}} d\mu = 1,$$

and define

$$F_{\text{APG}} = \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)}.$$

Let u_{APG} be the unique mean-zero solution of

$$\Delta_{\text{APG}} u_{\text{APG}} = F_{\text{APG}}.$$

Then

$$E_{\text{APG}} = \int_X |\nabla u_{\text{APG}}|^2 d\mu \leq \lambda_1(\Delta_{\text{APG}})^{-1} \|F_{\text{APG}}\|_{L^2(X)}^2.$$

Equivalently,

$$E_{\text{APG}} \leq \lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2.$$

Proof. Since X is compact and connected, the spectrum of the positive Laplacian satisfies

$$0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots.$$

The zero eigenspace consists exactly of constant functions. Since u_{APG} has mean zero, it is orthogonal to constants.

By elliptic spectral theory, expand F_{APG} in an orthonormal eigenbasis $\{\varphi_j\}_{j \geq 1}$ of mean-zero eigenfunctions:

$$F_{\text{APG}} = \sum_{j \geq 1} a_j \varphi_j, \quad \Delta_{\text{APG}} \varphi_j = \lambda_j \varphi_j.$$

The unique mean-zero solution is

$$u_{\text{APG}} = \sum_{j \geq 1} \frac{a_j}{\lambda_j} \varphi_j.$$

Therefore,

$$\|F_{\text{APG}}\|_2^2 = \sum_{j \geq 1} a_j^2.$$

Also,

$$E_{\text{APG}} = \int_X |\nabla u_{\text{APG}}|^2 d\mu = \int_X u_{\text{APG}} \Delta_{\text{APG}} u_{\text{APG}} d\mu.$$

Using the spectral expansion,

$$E_{\text{APG}} = \sum_{j \geq 1} \lambda_j \left(\frac{a_j}{\lambda_j} \right)^2 = \sum_{j \geq 1} \frac{a_j^2}{\lambda_j}.$$

Since $\lambda_j \geq \lambda_1$ for all $j \geq 1$,

$$E_{\text{APG}} = \sum_{j \geq 1} \frac{a_j^2}{\lambda_j} \leq \frac{1}{\lambda_1} \sum_{j \geq 1} a_j^2.$$

Hence

$$E_{\text{APG}} \leq \lambda_1^{-1} \|F_{\text{APG}}\|_2^2.$$

Substituting

$$F_{\text{APG}} = \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)}$$

gives

$$E_{\text{APG}} \leq \lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2.$$

This proves the theorem. □

7 Equivalent Green-Kernel Formulation

The same theorem may be expressed using the Green kernel $G_{\text{APG}}(x, y)$ of the mean-zero Laplacian. Since F_{APG} has mean zero,

$$u_{\text{APG}}(x) = \int_X G_{\text{APG}}(x, y) F_{\text{APG}}(y) d\mu(y).$$

Then

$$E_{\text{APG}} = \int_X \int_X F_{\text{APG}}(x) G_{\text{APG}}(x, y) F_{\text{APG}}(y) d\mu(x) d\mu(y).$$

The Spectral–Green Bridge Theorem states that the Green operator satisfies

$$\langle F_{\text{APG}}, G_{\text{APG}} F_{\text{APG}} \rangle \leq \lambda_1^{-1} \|F_{\text{APG}}\|_2^2.$$

This viewpoint is consistent with classical Green-kernel methods in elliptic spectral theory (Jost, 2017; Buser, 1992). In particular,

$$\|G_{\text{APG}}\|_{L_0^2(X) \rightarrow L_0^2(X)} = \lambda_1^{-1}.$$

This identifies the analytic heart of the APG VII Green-distortion problem.

8 Corollary: APG Spectral Stability

Corollary 8.1 (Spectral Stability of APG Energy). *If there exists a constant $\Lambda_N > 0$ such that*

$$\lambda_1(\Delta_{\text{APG}}) \geq \Lambda_N,$$

then

$$E_{\text{APG}} \leq \Lambda_N^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_2^2.$$

In particular, if Λ_N is bounded below uniformly over a chosen modular family, then APG energy is uniformly controlled by the L^2 -size of the mean-zero source.

Proof. The result follows directly from Theorem 6.1 by replacing $\lambda_1(\Delta_{\text{APG}})$ with its lower bound Λ_N . $\square$

9 Corollary: Spectral Conductor Criterion

Corollary 9.1 (Spectral Conductor Criterion). *Assume that for a modular level N and modular parametrization $\pi : X_0(N)^* \rightarrow E$, the APG source satisfies*

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_2^2 \leq C \deg(\pi) \log N$$

for an explicit constant $C > 0$. Then

$$E_{\text{APG}} \leq C \deg(\pi) \log N.$$

Proof. By Theorem 6.1,

$$E_{\text{APG}} \leq \lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_2^2.$$

The assumed spectral-conductor inequality gives

$$E_{\text{APG}} \leq C \deg(\pi) \log N.$$

This proves the corollary. $\square$

10 Conditional Link to Green-Energy Metric Distortion

Let M_{dist} denote the APG metric-distortion potential appearing in the APG–Arakelov comparison problem. The Spectral–Green Bridge does not automatically imply a bound for M_{dist} unless M_{dist} is identified with u_{APG} or is controlled by it. We therefore state the correct conditional form.

Proposition 10.1 (Conditional Green-Distortion Transfer). *Assume there exists a constant $C_0 > 0$ such that*

$$\int_X |\nabla M_{\text{dist}}|^2 d\mu \leq C_0 \int_X |\nabla u_{\text{APG}}|^2 d\mu.$$

If the Spectral Conductor Criterion holds, then

$$\int_X |\nabla M_{\text{dist}}|^2 d\mu \leq C_0 C \deg(\pi) \log N.$$

Proof. By the assumed comparison inequality,

$$\int_X |\nabla M_{\text{dist}}|^2 d\mu \leq C_0 E_{\text{APG}}.$$

By Corollary 9.1,

$$E_{\text{APG}} \leq C \deg(\pi) \log N.$$

Therefore,

$$\int_X |\nabla M_{\text{dist}}|^2 d\mu \leq C_0 C \deg(\pi) \log N.$$

This proves the proposition. $\square$

11 The APG VIII Breakthrough Perspective

The APG VII projection wall required control of Green-energy distortion. The present paper proves that Green-energy distortion is not an undefined obstruction. It is transformed into a spectrally measurable quantity. The obstruction is transformed into the following precise inequality:

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_2^2 \leq C \deg(\pi) \log N.$$

This is the APG VIII spectral-conductor problem.

The significance is conceptual and technical:

1. APG energy is no longer only a formal deformation quantity.
2. APG energy becomes a Green-energy quantity.
3. Green energy becomes a spectral quantity.
4. The remaining APG–Arakelov obstruction becomes a conductor-scale eigenvalue/source problem.
5. The full projection theorem is reduced to a precise analytic inequality.

This is a genuine strengthening of the APG program because it replaces a vague boundary condition with a theorem plus a clear target estimate.

12 Relation to Arakelov Intersection Theory

Arakelov geometry enriches algebraic intersection theory by including archimedean metric data (Arakelov, 1974; Faltings, 1984; Gillet & Soulé, 1990; Soulé, 1992). On arithmetic surfaces, changes in Hermitian metrics contribute analytic energy terms to arithmetic self-intersections. If a Hermitian metric is conformally varied by e^{-u} , then the associated arithmetic Chern self-intersection changes by a Dirichlet-energy contribution together with linear curvature terms.

In the APG setting, u_{APG} is the canonical mean-zero Poisson potential generated by the APG source. Therefore,

$$E_{\text{APG}} = \int_X |\nabla u_{\text{APG}}|^2 d\mu$$

is naturally positioned as the archimedean energy component of a metric variation. This is why the Spectral–Green Bridge is relevant to APG–Arakelov projection.

However, the exact comparison

$$E_{\text{APG}} \rightsquigarrow |\widehat{c}_1(\overline{\omega}_X)^2 - \deg(\pi)\widehat{c}_1(\overline{\omega}_E)^2|$$

requires additional arithmetic control over cusp terms, vertical fibers, and metric discrepancy. Those terms are not eliminated in this paper. Instead, this paper proves the analytic part required before those arithmetic estimates can be meaningfully attacked.

13 Numerical Illustration

Consider a normalized model with

$$\text{Vol}(X) = 1, \quad \lambda_1(\Delta_{\text{APG}}) = 0.25.$$

Suppose the APG source satisfies

$$\|\rho_{\text{APG}} - 1\|_2^2 = 0.04.$$

Then the Spectral–Green Bridge gives

$$E_{\text{APG}} \leq \frac{1}{0.25}(0.04) = 0.16.$$

If additionally

$$\deg(\pi) = 100, \quad N = 37, \quad C = 0.001,$$

then

$$C \deg(\pi) \log N = 0.001(100) \log(37) \approx 0.361.$$

Since

$$0.16 < 0.361,$$

the spectral-conductor criterion is satisfied in this model. This example is illustrative only. It does not prove the conductor inequality for all modular levels. It shows how APG VIII converts the analytic obstruction into a computable spectral test.

14 Discussion

The Spectral–Green Bridge Theorem is not a proof of Fermat’s Last Theorem, nor is it a substitute for the Ribet–Wiles modular method. Its role is internal to APG. It strengthens APG by proving that the APG source-energy mechanism has a rigorous spectral foundation.

The proof relies only on compactness, elliptic spectral theory, the mean-zero Poisson equation, and the positivity of the first nonzero Laplace eigenvalue. These are standard mathematical tools, but their integration into APG creates a new structural bridge:

$$\begin{aligned} \text{entropy/deformation source} &\longrightarrow \text{Poisson potential} \\ &\longrightarrow \text{Green energy} \longrightarrow \text{spectral conductor problem.} \end{aligned}$$

This is the correct level of analytical progress presently available. It is mathematically rigorous and reduces the principal APG VII Green-energy obstruction to a precise spectral-conductor problem, while avoiding unsupported claims regarding unresolved arithmetic consequences.

15 Open Problems

Open Problem 15.1 (Spectral-Conductor Bound). *Prove that*

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_2^2 \leq C \deg(\pi) \log N$$

for an explicit constant $C > 0$.

Open Problem 15.2 (Source Regularity). *Determine whether the APG source ρ_{APG} satisfies uniform Sobolev or L^∞ bounds over prime-chain modular levels.*

Open Problem 15.3 (Metric-Distortion Comparison). *Prove or disprove the existence of*

$C_0 > 0$ such that

$$\|\nabla M_{\text{dist}}\|_2^2 \leq C_0 \|\nabla u_{\text{APG}}\|_2^2.$$

Open Problem 15.4 (Cusp-Width Compatibility). *Relate weighted APG source concentration near cusps to classical cusp-width bounds on $X_0(N)$.*

Open Problem 15.5 (Vertical Fiber Control). *Connect the analytic spectral-conductor criterion to vertical component multiplicity estimates in regular arithmetic models.*

Open Problem 15.6 (Full APG–Arakelov Projection Theorem). *Determine whether the three boundary estimates from APG VII imply*

$$|\widehat{c}_1(\overline{\omega}_X)^2 - \deg(\pi)\widehat{c}_1(\overline{\omega}_E)^2| \leq C \deg(\pi) \log N.$$

16 Conclusion

This paper proves the APG Spectral–Green Bridge Theorem, establishing that APG deformation energy on a compactified modular curve is bounded by the inverse spectral gap times the squared L^2 -size of the mean-zero APG source. The result is unconditional once the APG metric and source are fixed smoothly on the compact Riemann surface.

The theorem advances APG VIII by transforming the APG VII Green-energy boundary obstruction into a precise spectral-conductor inequality. This is the main contribution of the paper. It does not prove the full APG–Arakelov Projection Theorem, but it provides a rigorous analytic foundation for the next stage of the APG program.

The correct future direction is therefore clear:

Prove the spectral-conductor inequality.

If that inequality is established with explicit constants, then one of the major conditional walls in APG VI–VII becomes an unconditional analytic theorem.

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